# Pooling Contracts between Venture Capital Institutions

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**Abstract**

Contracts between venture capitalists and entrepreneurs are generally startup specific. However, contracts between accelerators and venture capitalists, and between venture capitalists and limited partners usually take the form of pooling contracts. This article shows that pooling contracts between venture capital institutions emerge from information asymmetry and market power in the venture capital market. If the contract proposer reveals the expected outcome of the project after the investor's commitment as in the relationship between venture capitalists and limited partners, there will exist an optimal pooling mechanism in all economies, with a non-binding budget constraint for the proposer in cash-rich economies. Otherwise, if the contract proposer reveals the expected outcome of the project before the investor's commitment as in the relationship between accelerators and venture capitalists, the unique optimal mechanism will be pooling with a binding budget constraint for the proposer in all economies. This latter case is inefficient in cash-rich economies, rationalizing government intervention in the venture capital market through limited partnership.

**JEL Codes:** G24, G23, D82, D86

**Keywords:** Accelerators, Venture Capital, Limited Partners, Informed Principal, Mechanism Design, Blind-pool Contracts, Budget Constraints

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# 1 Introduction

Since the early 1980s, the venture capital market has consisted of venture capitalists (VCs), firms specialized in startup projects sourcing, financing and advising, themselves being funded by limited partners (LPs), which are mainly pension funds but also endowment funds, sovereign funds, insurance funds, funds of funds, etc. (Lerner and Nanda, 2020; Casamatta, 2003). Typically, most of their contracts are blind pool contracts (Lerner et al., 2022): they are agreed on through Limited Partnership Agreements (LPAs) before project sourcing. Indeed, LPs and VCs agree on an initial amount of callable investment called the committed capital; typically, VCs contribute from 1 to 5%. Then, VCs charge LPs according to a "2 and 20" fee structure: 2% of committed capital during the investment period and of net invested capital after the investment period; and 20% of net fund profits referred to as carried interest (Zeisberger et al., 2017). This contract is controversial since management fees may incentivize funds to grow at the expense of performance (Lerner and Nanda, 2020; Metrick and Yasuda, 2010). Consequently, LPs' blind commitment is suspected to be socially suboptimal (Lerner and Nanda, 2020), but inescapable due to lack of market power (Gourier et al., 2024; Lerner and Nanda, 2020) and since LPs co-investing on a deal-by-deal basis yields worse results than with blind pool deals (Fang et al., 2015). Axelson et al. (2009) already explain that private equity funds raise capital ex-ante so that managers are not incentivized to invest in bad deals. We ask why, in addition, the terms of the contract are the same for every deal.

Still, in the mid 2000s, with the introduction of Amazon Web Services (AWS) democratizing the use of the cloud for software solutions, the venture capital market evolved: software technologies became the main target, experimentation costs fell, VCs' sourcing and advising role eroded and ultimately, accelerators emerged as standard early-stage startup investors (Gonzalez-Uribe and Leatherbee, 2017; Ewens et al., 2018; Lerner and Nanda, 2020). Typically, accelerators propose fixed-term cohort-based programs that combine sourcing, financing and advising, culminating in a public pitch event, or "demo day", with potential investors (Cohen, 2013; Cohen and Hochberg, 2014; Hochberg, 2016). Many accelerator programs provide a small investment to their start-ups and receive an equity stake in the portfolio company in return, typically 5 to 7% (Hochberg, 2016). Accelerators also produce information on their start-ups. In the game-theoretic model of Zarei et al. (2022), screening is the service that matters most, and Kim and Wagman (2014) show that an accelerator chooses a portfolio smaller than the efficient one and may disclose only its positive signals to investors. We ask instead what an accelerator can credibly convey through its contracts: when it reveals its information before the investors commit, nothing beyond a pooling contract. Created in 2005, Y Combinator, founded by Paul Graham in the United States, is the first accelerator in the world. Initially, these were Y Combinator's deal terms: "Yes answers will include the valuation at which we're willing to invest. It will ordinarily be in the low to mid hundreds of thousands, but may occasionally be higher. Unless you've told us about specific hardware you need, the

total investment will be about \$6000n, where n is the number of people you'll be sending to Cambridge for the summer."[1] These deal terms were startup-specific. Pooling contracts started with Techstars, founded in 2007 (Cohen and Hochberg, 2014): "In exchange for the TechStars summer program, seed funding, advice, mentorship, connections, and investor demo day, TechStars receives a 5% equity stake in your new company. [...] You get \$5,000 per founder, up to 3 founders. So at most, you'll get \$15,000 for the summer."[2] Y Combinator's deal progressively became pooling afterwards.[3] Nowadays, Y Combinator complements its \$125K post-money Simple Agreement for Future Equity (SAFE) for 7% equity with a \$375K uncapped Most-Favored-Nation (MFN) SAFE, which converts at the best terms the startup later grants to seed investors.[4] This combination of fixed terms and an MFN SAFE is spreading across accelerators.[5] Dossou (2026) shows that, when the accelerator invests, the MFN clause gives it a higher profit than any fixed share, while its effect on the startups' probability of reaching Series A is not signed. The contract remains pooling. Every startup receives the same terms, whatever the accelerator's assessment of the project, and the MFN SAFE converts at the terms the seed investor sets.

The aim of this paper is twofold: we show how information asymmetry and market power explain the emergence of pooling contracts between institutions in the venture capital market; then, we demonstrate that the evolution of the venture capital market, documented by Ewens et al. (2018), has been so at an efficiency cost. We derive insights for rationalizing government intervention for supporting entrepreneurship. The paper builds on the informed-principal framework of Myerson (1983), Maskin and Tirole (1992), and Holmstrom and Myerson (1983). We solve the agency's problem with Lagrangian and optimal control methods (Guesnerie and Laffont, 1984; Kamien and Schwartz, 1991).

Indeed, we study contracts in which the first mover ("the agency") secures the followers' ("the investors") commitment to co-invest despite incentive frictions. When the principal reveals the expected outcome of the projects after the investors' commitment (as in the relationship between VCs and LPs), investors may tolerate losses on individual projects as long as they break even on average; when the principal reveals

[1] Y Combinator, "Summer Founders Program," 2005, `http://ycombinator.com/sfp.html`, archived January 11, 2006, at `https://web.archive.org/web/20060111165047/http://ycombinator.com/sfp.html`.

[2] TechStars, "Details," 2008, `http://www.techstars.org/details/`, archived January 17, 2008, at `https://web.archive.org/web/20080117052839/http://www.techstars.org/details/`.

[3] Paul Graham, comments on Hacker News, April 11, 2011, `https://news.ycombinator.com/item?id=2431519`, and February 9, 2013, `https://news.ycombinator.com/item?id=5191204`; Sam Altman, "The New Deal," Y Combinator Blog, April 22, 2014, `https://www.ycombinator.com/blog/the-new-deal`.

[4] Y Combinator, "The Y Combinator Standard Deal," `https://www.ycombinator.com/deal/`, accessed October 1, 2026.

[5] Techstars, "Investment Terms Update," April 17, 2025, `https://www.techstars.com/newsroom/investment-terms`; Entrepreneur First, "Updating Our Global Pre-Seed Investment Offer to up to \$250k," May 24, 2024, `https://www.joinef.com/posts/updating-our-global-pre-seed-investment-offer/`.

the expected outcome of the projects before the investors' commitment (as in the relationship between accelerators and VCs), investors may refuse to lose money on any project. We find that under either commitment regime, some pooling contracts are optimal for the agency. In particular, under *ex-post commitment*, when investors require incentive-compatible deal-by-deal profitable contracts, the unique optimal contract is pooling: the agency ensures that the investors are confident in the fact that the agency is not incentivized to misvalue any project because it could yield a negative utility for the investors. Pooling contracts, combined with a binding budget constraint on the agency, ensure that every financed project yields positive utility to investors. Since the agency invests the same amount in every financed project and spends its whole budget, its budget fixes the number of financed projects, and the agency picks the best ones. Under *ex-ante commitment*, the agency's budget constraint is not binding in general. In this case, the agency extracts all the investors' surplus and, when possible, funds all the profitable projects. But under ex-post commitment, there are always some profitable projects which remain unfunded.

The rest of the paper proceeds as follows. Section 2 describes the model and shows that pooling contracts follow from Myerson's inscrutability principle. Section 3 solves the benchmark in which investors share the agency's information. Section 4 characterizes the optimal contract under *ex-post commitment*, where investors refuse any project that loses money on its own. Section 5 does the same under *ex-ante commitment*, where investors only need to break even in expectation. Section 6 compares cash-poor and cash-rich economies. Section 7 concludes. All proofs are in the Appendix.

# 2 Model

## 2.1 Agents

Suppose there is no time preference and all the agents below are risk-neutral.

Let's introduce entrepreneurs without initial budget. Each entrepreneur carries a project which requires a sunk investment cost $I \in (0,1)$ and yields a stochastic outcome in $\{0,1\}$. 0 means failure and 1 means success. Without $I$, the project cannot succeed. An entrepreneur must hence collect at least $I$ from outside sponsors, who can split the investment burden. A project is characterized by its probability of success $\pi \in [0,1]$. Note that the probability of success is equal to the expected outcome. Each entrepreneur knows the expected outcome of his project. Let $f$ be the probability density function of $\pi$, and let $F$ denote its cumulative distribution function.

Potential sponsors are investors with an aggregate budget $B_i \geq 0$. The investors know the distribution of $\pi$. In addition, they form a competitive market: they strategically

underbid one another in order to secure investment opportunities. We represent them by a single price-taking investor.

Financing can also be obtained from an agency with an initial budget $B_a \geq 0$. For each project, the agency knows the exact value of its expected outcome. An *investment allocation* $\sigma$ specifies, for each project $\pi$, the agency's contribution to the investment cost $I$. The investor is expected to complete the remaining $\sigma_i(\pi)$. A *share allocation* $\rho$ specifies, for each project $\pi$, the outcome $\rho(\pi)\pi$ obtained by the agency. The investor receives the remaining $\rho_i(\pi)\pi$. Formally:

$$\sigma_i(\pi) = \begin{cases} I - \sigma(\pi) & \text{if } \sigma(\pi) > 0 \\ 0 & \text{otherwise} \end{cases}$$

and:

$$\rho_i(\pi) = \begin{cases} 1 - \rho(\pi) & \text{if } \rho(\pi) > 0 \\ 0 & \text{otherwise} \end{cases}$$

The agency signals projects to investors by investing in them. Thereby, we combine the certification and financing channels proposed by Howell (2017).

## 2.2 Definitions and assumptions

A project's *net present value* is defined as the difference between its expected outcome and its investment cost. For notational simplicity, the *market net present value*, which corresponds to the outside option's net present value, is taken to be 0. Then, a project is *worth financing* if its net present value is positive. A project is *financed* if it receives the required investment cost.

(A1): $\mathbb{E}(\pi) - I < 0 < 1 - I$ and $f$ is log-concave

The first part of (A1) means that the average project is not worth financing but that the best project is worth financing. The second part holds for the uniform, normal, logistic and exponential distributions, and implies that the hazard rate $f/(1-F)$ is increasing.

(A2): $B_a < I(1 - F(I))$ and $B_a + B_i \geq I(1 - F(I))$

The first part of $(A2)$ means that the agency cannot finance all worth financing projects alone: it needs outside investors. This is the relevant case in practice, since even the largest VCs and accelerators cannot finance all the projects alone. The second part means that together the agency and the investor can finance all projects.

(A3): The agency is profit-maximizing

The agency's payoff is:

$$U_a(\pi) = \rho(\pi)\pi - \sigma(\pi) \quad \text{for each } \pi$$

The investors' payoff is:

$$U_i(\pi) = (1 - \rho(\pi))\pi - (I - \sigma(\pi)) \quad \text{for each } \pi$$

The investment strategies must be budget balanced both for the agency and the investor:

$$B_a \geq \int \sigma(\pi) f(\pi) d\pi \tag{$BC_a$}$$

$$B_i \geq \int \sigma_i(\pi) f(\pi) d\pi \tag{$BC_i$}$$

We have assumed here that the budgets $B_a$ and $B_i$ are common knowledge.

# 3 Benchmark

We first show that even without information asymmetry, the first mover extracts all the followers' surplus. We will also introduce conceptual notions for the general resolution.

**Game 1** (Benchmark)**.**

$t = 0$ *Nature draws* $\pi$

$t = 1$ *The agency and the investors learn* $\pi$

$t = 2$ *The agency proposes* $(\rho(\pi), \sigma(\pi))$

$t = 3$ *Investors accept or refuse*

$t = 4$ *The agency executes* $(\rho(\pi), \sigma(\pi))$

$t = 5$ *Payoffs are realized*

**Proposition 1.** *Any budget balanced contract which finances all the worth financing projects by binding the investor's participation constraint on each project is optimal.*

*Proof.* See Appendix. □

Indeed, since the agency is profit-maximizing, the investor's rationality constraint is binding for each project. Otherwise, since the investor market is competitive, the agency could contract with a cheaper investor, thereby making more profit. Thus, the agency's optimization problem directly reduces to a welfare maximization problem under budget constraints. This proposition means that when there is no asymmetric information, the agency extracts all the surplus from the investor project-wise, thanks to its first mover advantage.

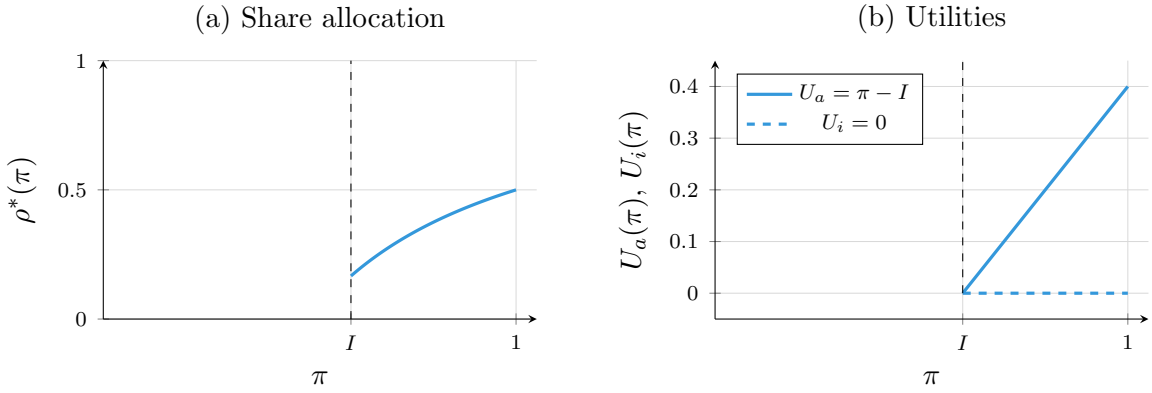



Figure 1: Informed investors under the benchmark. Parameters: $\pi \sim U[0,1]$, $I = 0.6$, $\sigma^* = 0.1$. (a) The share $\rho^*(\pi) = 1 - (I - \sigma^*)/\pi$ is increasing in $\pi$. (b) The agency captures the full surplus $\pi - I$ and the investor earns zero.

# 4 Ex-post commitment investors

**Game 2** (Ex-post commitment)**.**

$t = 0$ *Nature draws* $\pi$

$t = 1$ *The agency learns* $\pi$

$t = 2$ *The agency proposes* $(\rho, \sigma)$

$t = 3$ *The agency reveals* $\pi$

$t = 4$ *The investors accept or refuse*

$t = 5$ *The agency executes* $(\rho(\pi), \sigma(\pi))$

$t = 6$ *Payoffs are realized*

The agency, here mapping to the accelerator, learns the expected outcome of the project before proposing the contract. Indeed, we want to establish that even if the

accelerator knew the exact value of the project, it would rather propose a pooling contract. The third stage here corresponds to the "demo day" where the entrepreneurs publicly pitch their projects to potential follow-on investors. Investors step in only if they break even on each specific displayed project. Formally, under ex-post commitment investors, the individual-rationality constraint must hold pointwise, on every financed project:

$$\forall \pi, \quad (1-\rho(\pi))\pi - (I - \sigma(\pi)) \geq 0 \qquad (IR_{i,p})$$

| Project | $\pi = 1/2$ | $\pi = 1/2$ | $\pi = 3/4$ | $\pi = 1$ |
|---|---|---|---|---|
| *(a) A non-implementable separable contract* | | | | |
| $(\rho, \sigma)$ | (0.1, 0.05) | (0.1, 0.05) | (0.4, 0.05) | (0.6, 0.10) |
| Agency $U_a$ | 0 | 0 | 0.25 | 0.50 |
| Investor $U_i$ | 0 | 0 | 0 | 0 |
| *($a_{bis}$) The agency reports the second project as $\pi = 1$* | | | | |
| $(\rho, \sigma)$ | (0.1, 0.05) | (0.6, 0.10) | (0.4, 0.05) | (0.6, 0.10) |
| Agency $U_a$ | 0 | 0.20 | 0.25 | 0.50 |
| Investor $U_i$ | 0 | $-0.20$ | 0 | 0 |
| *(b) The optimal contract* | | | | |
| $(\rho, \sigma)$ | — | — | (0.6, 0.20) | (0.6, 0.20) |
| Agency $U_a$ | — | — | 0.25 | 0.40 |
| Investor $U_i$ | — | — | 0 | 0.10 |

Table 1: Four projects, $I = 1/2$, $B_a = 0.4$. (a) would leave every investor indifferent but cannot be implemented since ($a_{bis}$) shows the agency is better off by announcing a $\pi = 1/2$ project as $\pi = 1$, which pushes that investor to $-0.20$. (b) is the optimum of Proposition 2.

Under ex-post commitment, the optimal allocation is pooling, the agency spends its whole budget, and some positive-NPV projects are not financed.

**Proposition 2.** *Under ex-post commitment investors, the optimal allocation is pooling. Any optimal allocation $(\rho^*, \sigma^*, \pi^*_{\min})$ satisfies:*

1. *Some positive-NPV projects are excluded: $\pi^*_{\min} > I$.*
2. *The agency saturates its budget:*

$$\sigma^* = \frac{B_a}{\int_{\pi^*_{\min}}^1 f(\pi)\, d\pi}.$$

3. *The share is determined by the threshold:*

$$\rho^* = 1 - \frac{I - \sigma^*}{\pi^*_{\min}}.$$

4. *If $\sigma^* < I$, the optimal threshold $\pi^*_{\min}$ satisfies the first-order condition:*

$$(\pi^*_{\min} - I) f(\pi^*_{\min}) = \frac{\int_{\pi^*_{\min}}^{1} (\pi - \pi^*_{\min}) f(\pi)\, d\pi}{\pi^*_{\min} \int_{\pi^*_{\min}}^{1} f(\pi)\, d\pi} \cdot \sigma^* \cdot f(\pi^*_{\min}) + \frac{I - \sigma^*}{(\pi^*_{\min})^2} \int_{\pi^*_{\min}}^{1} \pi f(\pi)\, d\pi.$$

   *The right-hand side is strictly positive, hence $\pi^*_{\min} > I$.*

5. *Otherwise $\sigma^* = I$, $\rho^* = 1$ and $\pi^*_{\min} = (1 - F)^{-1}(B_a/I)$, and the agency finances alone every project above the threshold.*

*Proof.* See Appendix. □

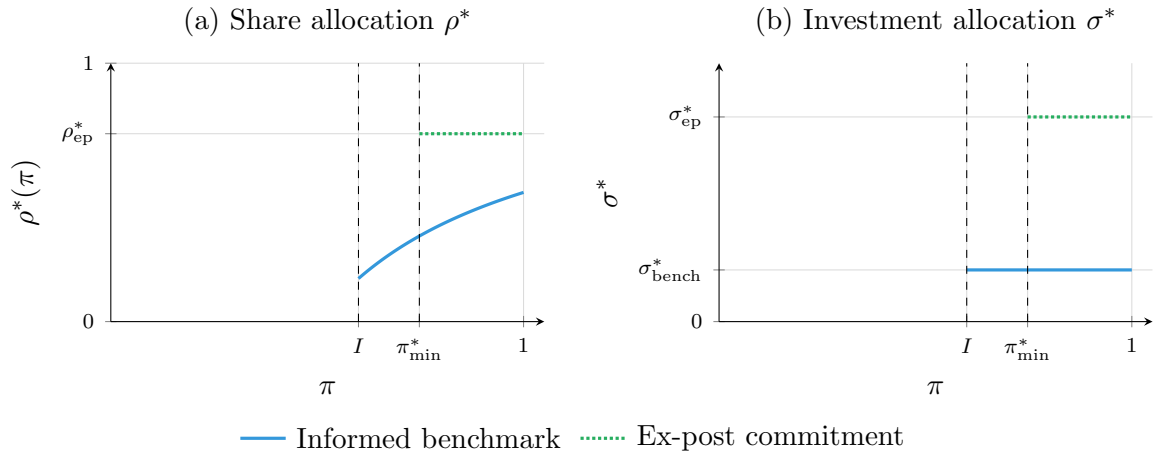



Figure 2: Informed benchmark vs. ex-post commitment regime. Parameters: $\pi \sim U[0,1]$, $I = 0.6$, $B_a = 0.1$, $B_i = 0.2$. Under the informed benchmark, $\rho^*(\pi) = 1 - (I - \sigma^*)/\pi$ for each project. Under ex-post commitment, the optimal allocation is pooling and the threshold rises above $I$. The agency finances fewer projects but invests more per financed project.

The agency maximizes its profit under its own pointwise incentive compatibility constraint (the utility it gets from applying to a project its own contract is greater than the one it would get by applying any other project's contract), the investors' pointwise participation constraint, and budget constraints. The intuition is that to be credible, the agency must apply either a uniform contract or contracts such that investment and share splits are higher for the better projects. When optimizing, the latter proves incompatible with the investors' participation constraint. A strictly increasing allocation always leaves room for investors' negative utility on a given project through exchanging contracts. The unique credible safety measure is pooling contracts together. The agency's budget constraint binds. With the same investment $\sigma^*$ in every financed project, the budget fixes the number of financed projects, and since $U_a(\pi) = \rho^*\pi - \sigma^*$ increases in $\pi$, the agency fills them with the best projects.

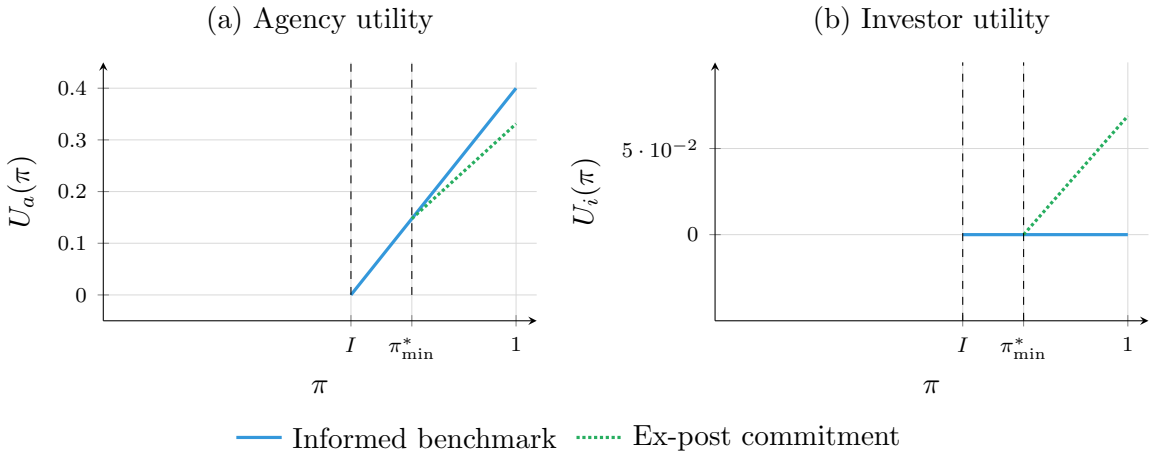



Figure 3: Utilities per project: informed benchmark vs. ex-post commitment regime. Same parameters as Figure 2. (a) Agency utility: the benchmark captures $\pi - I$; ex-post commitment captures less on a smaller range. (b) Investor utility: zero under the benchmark, strictly positive under ex-post commitment.

# 5 Ex-ante commitment investors

**Game 3** (Ex-ante commitment)**.**

$t = 0$ *Nature draws* $\pi$

$t = 1$ *The agency learns* $\pi$

$t = 2$ *The agency proposes* $(\rho, \sigma)$

$t = 3$ *The investors accept or refuse*

$t = 4$ *The agency reveals* $\pi$

$t = 5$ *The agency executes* $(\rho(\pi), \sigma(\pi))$

$t = 6$ *Payoffs are realized*

Here, at stages 2 and 3, the agency and the investors, mapping respectively to VCs and LPs, establish the LPA (Limited Partnership Agreement) establishing rules for the investment and return splits. Afterwards, the VC makes capital calls to draw down amounts of committed capital from the LPs. It is only at this stage that portfolio projects can be known. Since LPs' decision is made on the whole portfolio (and the outside option is normalized to zero), their participation is conditioned on a positive gain in expectation[6]:

$$\int \big[\rho_i(\pi)\pi - \sigma_i(\pi)\big] f(\pi)\, d\pi \geq 0 \qquad (IR_i)$$

[6]Maskin and Tirole (1992)

As in the benchmark, the agency extracts the full investor surplus. The pooling allocation below reaches the total surplus that the budgets allow, so no incentive-compatible allocation does better.

**Proposition 3.** *Under ex-ante commitment investors, a pooling allocation* $(\rho^*, \sigma^*)$ *with financing threshold* $\pi_{\min}$ *is an optimal solution, characterized by:*

1. *The financing threshold is* $\pi^*_{\min} = I$*: all positive-NPV projects are financed.*
2. *The agency's share is:*

$$\rho^* = 1 - \frac{(1 - F(I))(I - \sigma^*)}{\int_I^1 \pi f(\pi)\, d\pi}$$

3. *The agency's investment share belongs to:*

$$\sigma^* \in \left[ I - \frac{B_i}{1 - F(I)},\ \frac{B_a}{1 - F(I)} \right] \cap (0, I]$$

*The cash-poor case* $(B_a + B_i < I(1 - F(I)))$ *is also treated in the Appendix.*

*Proof.* See Appendix. □

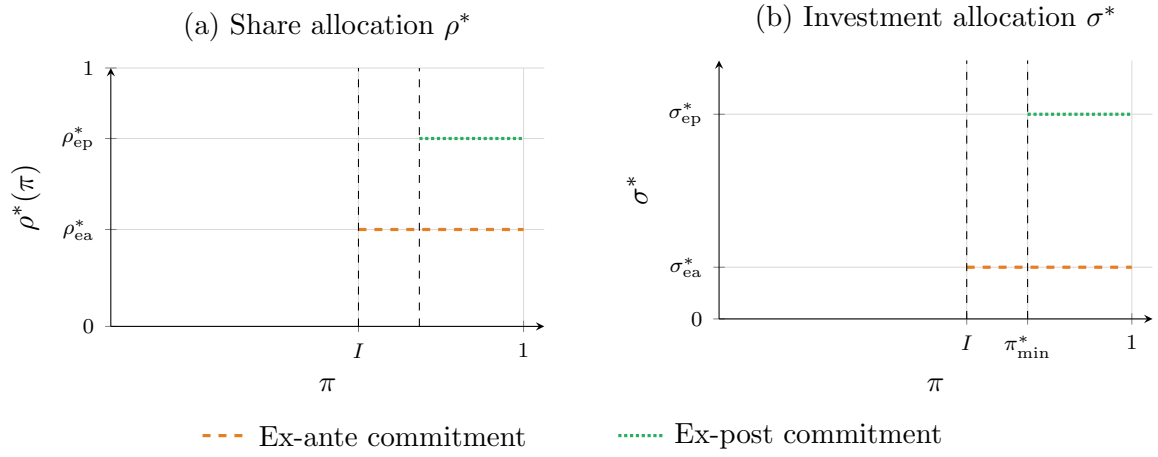



Figure 4: Optimal allocation under ex-ante vs. ex-post commitment investors. Same parameters as Figure 2. The ex-ante commitment regime finances all positive-NPV projects ($\pi^*_{\min} = I$); the ex-post commitment regime excludes the interval $[I, \pi^{\text{ep}}_{\min})$ and invests more per financed project.

Since the LPs' average surplus must be positive (better than the outside option), the VC should minimize it while remaining credible. Uniform contracts ensure credibility (since the contracts are the same for all the projects, there is no interest for exchanging

any contracts) and proposing the uniform contract such that the investors' utility in the average project is zero guarantees a maximal utility for the VC. The VC just has to invest enough budget in each project so that the LPs are able to complete the remaining for financing all the NPV-positive projects. Indeed, since the VC extracts all the surplus, the more projects are financed, the more the VC is profitable.

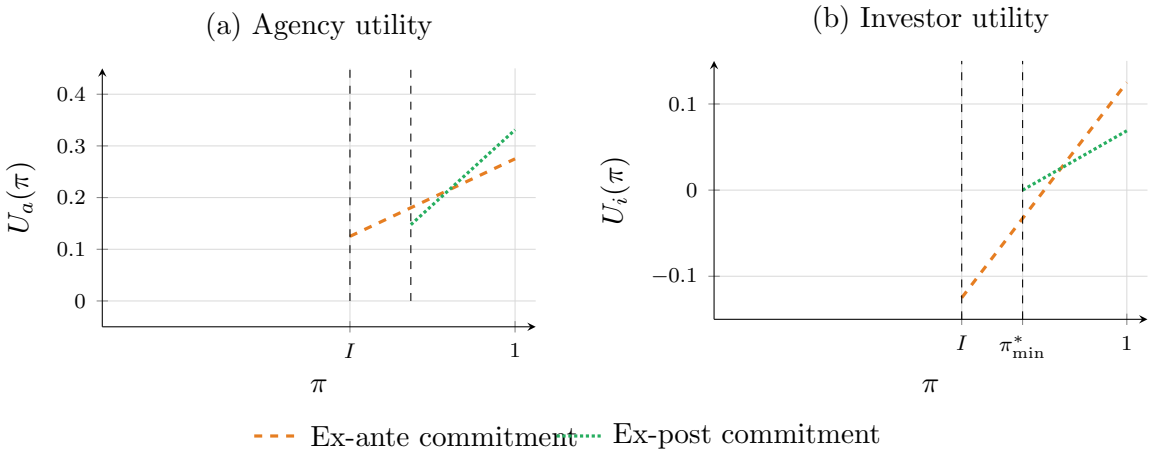



Figure 5: Utilities under the two commitment regimes. Same parameters. (a) Agency utility: the ex-ante commitment regime yields higher expected profit. (b) Investor utility: zero in expectation under ex-ante commitment (negative for low-$\pi$ projects, positive for high-$\pi$ projects), and strictly positive for every $\pi \geq \pi_{\min}$ under ex-post commitment.

# 6 Cash-poor vs. cash-rich economies

So far, the second part of (A2) has ensured that the two budgets together can finance every positive-NPV project. We now drop it and describe the whole $(B_a, B_i)$ plane. In addition, until now, we have analyzed the ex-ante commitment and the ex-post commitment regimes as describing two separate worlds. But indeed, the ex-post commitment regime describes the post-AWS introduction world where accelerators have emerged as canonical early-stage investors, while the ex-ante commitment regime describes the pre-AWS introduction world. The budget analysis will allow us to make predictions of the effect of this transition on the efficiency of the venture capital market.

Let $\pi^{\text{ep}}_{\min}(B_a)$ denote the optimal threshold of Proposition 2 when $(BC_i)$ is slack.

**Proposition 4.** *In the $(B_a, B_i)$ plane, we can define three regions:*

1. ***Severely cash-poor economies:***
   - $B_a + B_i < I(1 - F(\pi^{\text{ep}}_{\min}(B_a)))$

- *Both budgets bind, and both financing thresholds are equal to* $(1-F)^{-1}\left(\frac{B_a+B_i}{I}\right)$.

2. ***Moderately cash-poor economies:***

- $I(1-F(\pi^{\text{ep}}_{\min}(B_a))) \leq B_a + B_i < I(1-F(I))$
- *Under ex-ante commitment, both budgets bind and the financing threshold is* $(1-F)^{-1}\left(\frac{B_a+B_i}{I}\right)$.
- *Under ex-post commitment, only the agency's budget binds and the financing threshold is* $\pi^{\text{ep}}_{\min}(B_a)$*, above* $(1-F)^{-1}\left(\frac{B_a+B_i}{I}\right)$.

3. ***Cash-rich economies:***

- $B_a + B_i \geq I(1-F(I))$
- *Under ex-ante commitment, no budget binds and the financing threshold is* $I$*: every positive-NPV project is financed.*
- *Under ex-post commitment, only the agency's budget binds and the financing threshold is* $\pi^{\text{ep}}_{\min}(B_a)$*, above* $I$.

In severely cash-poor economies, we don't expect any efficiency loss from the regime transition, while in both moderately cash-poor economies and cash-rich economies, there is an efficiency loss from the regime transition.

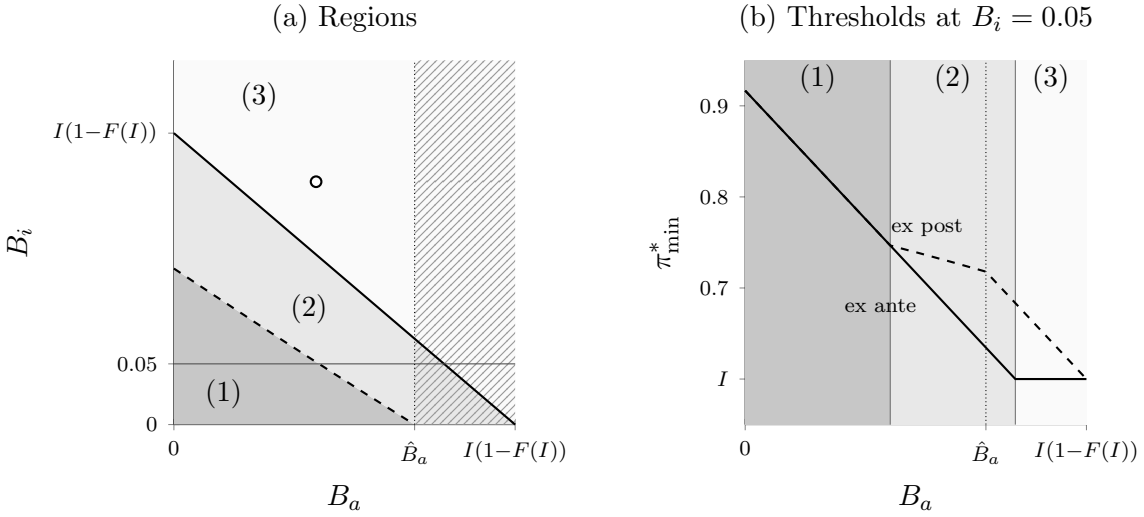



Figure 6: Budget regions ($f$ uniform, $I = 0.6$). Panel (a): regions in the $(B_a, B_i)$ plane. Panel (b): optimal thresholds at $B_i = 0.05$. Right of the dotted line, the ex-post commitment agency finances alone. In region (1), both regimes share the same threshold; in regions (2) and (3), the ex-post commitment threshold is higher.

# 7 Conclusion

## 7.1 Discussion

There are some key hypotheses behind our results which are worth mentioning. We model market power and information asymmetry with an informed monopolistic principal versus non-informed competitive agents. In particular, VCs are modelled as informed in their relationship with LPs, and as uninformed in their relationship with accelerators. Indeed, this is coherent with the story of the evolution of the venture capital market (Ewens et al., 2018), which is that the sourcing role has been transferred from VCs to accelerators. In their relationship with LPs, there is evidence that top-performing VCs have market power (Lerner and Nanda, 2020; Gourier et al., 2024). One may have preferred oligopolistic structures or partial information levels but our aim was to capture the essence of the relationships, which explains why these hypotheses are sufficient for our model.

In addition, the investment cost $I$ is the same for all the projects. In the real world, there is heterogeneity: different early-stage projects in a VC portfolio or in an accelerator cohort do not necessarily have the same investment need. But, this heterogeneity is in fact captured by the distribution of success probabilities. It would indeed be equivalent to say that the projects are costless but that their expected outcomes $\pi_I := \pi - I$ follow a distribution $f$. Similarly, one could argue that investment amounts may increase success probabilities. This is why we choose to normalize the investment cost: the investment need being fixed, what is the heterogeneity of success probabilities? This makes projects comparable.

Moreover, in our model, entrepreneurs are not active agents. They are only intermediaries between venture capital institutions who do not exert any effort or need any compensation. We could solve it by conditioning the non-zero project success probability to the entrepreneur effort exercise. But, since we want to focus on the relationships between venture capital institutions, we consider the entrepreneur is uninformed on the expected value of the project and thus, we incorporate his compensation in the investment cost. Admittedly, the model structure could be levelled up to take moral hazard into account.

Finally, we do not explain why the accelerator model is structured as an ex-post commitment model rather than as an ex-ante commitment model and leave it for further research.

## 7.2 Implications for practice and policy

Our paper provides a rationale for pooling contracts in the venture capital world. These contracts are free from information that the informed principal could provide and monetize as an expert, and this is why the ubiquity of pooling contracts

is counterintuitive. In a word, pooling contracts emerge as a solution for credibly communicating the deal fairness to priorly uninformed investors.

In addition, the paper identifies an efficiency cost from the transition the venture capital industry has made with the emergence of accelerators as canonical early-stage investors. Indeed, before the democratization of cloud services responsible for this transition, VCs were responsible for sourcing and advising. Thus, they were the informed principal. In their relationship with the LPs, our model shows that VCs' incentives were aligned with social welfare: the more projects are financed, the more the VC is profitable. This situation was socially optimal.

The structure of the accelerator model introduces an inefficiency. Indeed, the competence for sourcing and advising has been transferred to accelerators, but VCs still negotiate deal by deal (and not blindly commit to financing a whole batch of startups). Thereby, the informed principal, here the accelerator, reveals its private information before the agents' commitment, which introduces a social inefficiency.

This last argument can justify why governments should support entrepreneurship. Indeed, due to the venture capital market regime transition, the focus has been turned to the more profitable (or equivalently less costly) projects (in the software sector) at the peril of less profitable projects (in the hardware sector for example). We document an information structure distortion in the venture capital market, responsible for the capital rationing for projects perceived as less profitable: since the accelerator compensation is conditioned on the project success in raising VC capital, startup selection is harder and concentrated on projects on which evidence on success probability can be quickly (and less costly) made, and demonstrated to follow-on investors.

Last, our budget analysis is useful in terms of policy. In rich economies, we would not recommend that governments fund accelerators to cope with the venture capital market inefficiency. Consistent with our results, an efficient public policy could be to finance skilled venture capitalists specialized in specific fields usually perceived as less profitable because of the market pressure, as a limited partner so that they avoid the temptation of spraying capital on accelerator-selected startups. In poor economies, these considerations are irrelevant since what is really binding is the budget available. The money will in any case have to go first to most obviously profitable projects through accelerator or VC sourcing.

# 8 Appendix

## 8.1 Proof of Proposition 1 (informed investors).

**Lemma 1.** *No project with $\pi < I$ is financed in any optimal mechanism.*

*Proof.* If $\pi < I$, then $\pi - I < 0$: the project has negative expected net value. The investor's payoff from financing it is $(1 - \rho(\pi))\pi - (I - \sigma(\pi)) \leq \pi - I < 0$ whenever $\sigma(\pi) \leq \rho(\pi)\pi$ (i.e. the agency does not subsidize the investor beyond the project's return). Thus $(IR_{i,p})$ cannot be satisfied unless the agency transfers more than the project generates, which would also make the agency's payoff negative. Since both parties lose, no rational mechanism finances projects below $I$. □

**Step 1:** The investor's rationality constraint binds

Since the agency is profit-maximizing, the investor's rationality constraint is binding for each project. Otherwise, since the investor market is competitive, the agency could contract with a cheaper investor, thereby making more profit:

$$1 - \rho^*(\pi) > \frac{I - \sigma^*(\pi)}{\pi} \Rightarrow \exists \rho^{\#}(\pi) \in \Big(\rho^*(\pi), 1 - \frac{I - \sigma^*(\pi)}{\pi}\Big),$$
$$\int_{\pi^*_{\min}}^{1} (\rho^*(\pi)\pi - \sigma^*(\pi)) f(\pi)\, d\pi < \int_{\pi^*_{\min}}^{1} (\rho^{\#}(\pi)\pi - \sigma^*(\pi)) f(\pi)\, d\pi$$

and $(\sigma^*, \rho^{\#})$ satisfies $(BC_a)$, $(BC_i)$, and $(IR_i)$, a contradiction. Thus:

$$\rho^*(\pi) = 1 - \frac{I - \sigma^*(\pi)}{\pi}$$

**Step 2:** Reducing the agency's optimization problem

The agency's optimization problem reduces to:

$$\max_{(\pi_{\min}, \sigma, \rho)} \int_{\pi_{\min}}^{1} \Big(\Big(1 - \frac{I - \sigma(\pi)}{\pi}\Big)\pi - \sigma(\pi)\Big) f(\pi)\, d\pi$$
$$= \max_{(\pi_{\min}, \sigma, \rho)} \int_{\pi_{\min}}^{1} (\pi - I) f(\pi)\, d\pi$$

and:

$$\rho(\pi) = 1 - \frac{I - \sigma(\pi)}{\pi}$$

## 8.2 Proof of Proposition 3 (ex-ante commitment investors).

**Step 1:** The investor's rationality constraint binds in expectation.

Since the agency is profit-maximizing, $(IR_i)$ binds in expectation. Otherwise, the agency could reduce the investor's share and strictly increase its profit: if $(IR_i)$ were slack, there would exist $\hat{\rho}$ such that $(\sigma^*, \hat{\rho})$ remains feasible and yields a higher agency payoff, a contradiction. Hence:

$$(1-\rho^*)\,\frac{\int_{\pi^*_{\min}}^{1} \pi f(\pi)\,d\pi}{\int_{\pi^*_{\min}}^{1} f(\pi)\,d\pi} = I - \sigma^*.$$

**Step 2:** Reducing the optimization problem.

With the binding $(IR_i)$, the agency's objective becomes

$$\max_{\sigma,\pi_{\min}} \int_{\pi_{\min}}^{1} \Big(\Big(1 - \tfrac{(I-\sigma)(1-F(\pi_{\min}))}{\int_{\pi_{\min}}^{1} \pi f(\pi)\,d\pi}\Big)\pi - \sigma\Big) f(\pi)\,d\pi$$

under $(BC_a)$ and $(BC_i)$. Any allocation satisfying IC and the binding $(IR_i)$ delivers the same welfare, so the contract form is not uniquely determined.

**Step 3:** First-order conditions.

Expanding the derivative with respect to $\pi^*_{\min}$ by the Leibniz rule and the quotient rule yields

$$\frac{\partial \mathbb{E}[\pi \mid \pi \geq \pi_{\min}]}{\partial \pi_{\min}} = \frac{f(\pi_{\min})\big(\int_{\pi_{\min}}^{1} \pi f(\pi)\,d\pi - \pi_{\min}(1-F(\pi_{\min}))\big)}{(1-F(\pi_{\min}))^2}.$$

After substitution and simplification, the first-order condition on $\pi^*_{\min}$ reduces to $(-\pi^*_{\min} + I) f(\pi^*_{\min}) = 0$. Since $f(\pi^*_{\min}) > 0$, we obtain $\pi^*_{\min} = I$: all positive-NPV projects are financed.

The first-order condition with respect to $\sigma^*$ reduces to the identity $\mathbb{E}[\pi \mid \pi \geq \pi^*_{\min}] = \int_{\pi^*_{\min}}^{1} \pi f(\pi)\,d\pi / (1 - F(\pi^*_{\min}))$, which is true by definition. Hence the unconstrained first-order conditions do not pin down $\sigma^*$, and the budget constraints determine it.

**Step 4:** Budget case analysis under (A2).

The Lagrangian

$$\begin{aligned} L = &-\int_{\pi_{\min}}^{1} \Big(\Big(1 - \tfrac{(I-\sigma)(1-F(\pi_{\min}))}{\int_{\pi_{\min}}^{1} \pi f(\pi)\,d\pi}\Big)\pi - \sigma\Big) f\,d\pi \\ &+ \lambda_1\Big(\sigma\int_{\pi_{\min}}^{1} f - B_a\Big) + \lambda_2\Big((I-\sigma)\int_{\pi_{\min}}^{1} f - B_i\Big) \end{aligned}$$

gives the stationarity conditions

$$\frac{\partial L}{\partial \sigma} = (-\lambda_1 + \lambda_2)(1 - F(\pi_{\min})) = 0,$$
$$\frac{\partial L}{\partial \pi_{\min}} = \big((\pi_{\min} - I) - \lambda_1 \sigma - \lambda_2 (I - \sigma)\big) f(\pi_{\min}) = 0.$$

Hence $\lambda_1^* = \lambda_2^*$ and $\pi_{\min}^* = I + \lambda_1^* I$. Under (A2), $B_a + B_i \geq I(1 - F(I))$, so both multipliers vanish, $\pi_{\min}^* = I$, and the investment share satisfies

$$\sigma^* \in \Big] \tfrac{B_a}{1-F(I)},\ I - \tfrac{B_i}{1-F(I)} \Big[ \cap\, (0, I].$$

Any allocation in this interval satisfying IC and the binding $(IR_i)$ is optimal. □

## 8.3 Proof of Proposition 2 (ex-post commitment investors).

**Part I: pooling is optimal.**

We first write the incentive-compatibility constraint as two differential conditions, which serve as the state equations of the optimal control problem.

**Lemma 2.** *The agency's incentive-compatibility constraint is equivalent to its first- and second-order conditions:*

$$(IC_a) \Leftrightarrow \begin{cases} \dfrac{d\sigma(\pi)}{d\pi} = \pi \dfrac{d\rho(\pi)}{d\pi} & (IC_a^{(1)}) \\ \dfrac{d\rho(\pi)}{d\pi} \geq 0 & (IC_a^{(2)}) \end{cases}$$

*Proof.* ($\Rightarrow$) The first-order condition is

$$\frac{\partial V}{\partial \hat{\pi}}(\pi, \pi) = 0 \Leftrightarrow \frac{d\sigma(\pi)}{d\pi} = \pi \frac{d\rho(\pi)}{d\pi}.$$

Differentiating,

$$\frac{d^2\sigma(\pi)}{d\pi^2} = \frac{d\rho(\pi)}{d\pi} + \pi \frac{d^2\rho(\pi)}{d\pi^2},$$

and the second-order condition $\frac{d^2\sigma(\pi)}{d\pi^2} \geq \pi \frac{d^2\rho(\pi)}{d\pi^2}$ yields $\frac{d\rho(\pi)}{d\pi} \geq 0$ by substitution.

($\Leftarrow$) Consider

$$\frac{\partial V}{\partial \hat{\pi}}(\pi, \hat{\pi}) = \frac{\partial(\pi \rho(\hat{\pi}))}{\partial \rho} \frac{d\rho(\hat{\pi})}{d\pi} - \frac{d\sigma(\hat{\pi})}{d\pi}.$$

Writing the first-order condition at $\hat{\pi}$ gives $\frac{\partial(\hat{\pi}\rho(\hat{\pi}))}{\partial \rho} \frac{d\rho(\hat{\pi})}{d\pi} = \frac{d\sigma(\hat{\pi})}{d\pi}$, hence

$$\frac{\partial V(\pi, \hat{\pi})}{\partial \hat{\pi}} = \Big( \frac{\partial(\pi \rho(\hat{\pi}))}{\partial \rho} - \frac{\partial(\hat{\pi}\rho(\hat{\pi}))}{\partial \rho} \Big) \frac{d\rho(\hat{\pi})}{d\pi}.$$

By the Mean Value Theorem, the sign of the right-hand side is that of $\frac{\partial^2(\pi^*\rho(\hat{\pi}))}{\partial\rho\partial\pi}(\pi - \hat{\pi})\frac{d\rho(\hat{\pi})}{d\pi}$ for some $\pi^*$ between $\pi$ and $\hat{\pi}$. Since $\frac{\partial^2(\pi^*\rho(\hat{\pi}))}{\partial\rho\partial\pi} = 1$, the second-order condition implies that $\hat{\pi} \mapsto V(\pi, \hat{\pi})$ is increasing until $\hat{\pi} = \pi$ and decreasing afterwards, which concludes the proof. □

Using Lemma 2, the agency's program can be rewritten as the following optimal control problem:

$$\max_{(\pi_{\min},\sigma,\rho)} \int_{\pi_{\min}}^{1} (\rho(\pi)\pi - \sigma(\pi))f(\pi)\, d\pi$$

under:

$$\begin{aligned}
&B_a - \int_{\pi_{\min}}^{1} \sigma(\pi)f(\pi)\, d\pi \geq 0 && (BC_a)\\
&B_i - \int_{\pi_{\min}}^{1} (I - \sigma(\pi))f(\pi)\, d\pi \geq 0 && (BC_i)\\
&(1 - \rho(\pi))\pi - (I - \sigma(\pi)) \geq 0 && (IR_{i,p})\\
&\rho'(\pi) = u(\pi), \quad \sigma'(\pi) = \pi u(\pi) && (IC_a^{(1)})\\
&u(\pi) \geq 0 && (IC_a^{(2)})\\
&\sigma(\pi_{\min}), \rho(\pi_{\min}), \sigma(1), \rho(1) \text{ free} && (LC)
\end{aligned}$$

**Step 1:** Applying Pontryagin's Maximum Principle

The Hamiltonian is:

$$\begin{aligned}
H(\pi) =&(\rho(\pi)\pi - (1 + (\lambda_a - \lambda_i))\sigma(\pi))f(\pi)\\
&+\lambda_1(\pi)u(\pi) + \lambda_2(\pi)\pi u(\pi)\\
&+\eta_1(\pi)u(\pi) + \eta_2(\pi)(\pi - I - \rho(\pi)\pi + \sigma(\pi))
\end{aligned}$$

- Optimality condition

$$\frac{\partial H}{\partial u} = 0 \Leftrightarrow \lambda_1(\pi) + \lambda_2(\pi)\pi + \eta_1(\pi) = 0$$

- Multiplier conditions

$$\begin{aligned}
\lambda_1'(\pi) &= -\frac{\partial H}{\partial \rho} = \eta_2(\pi)\pi - \pi f(\pi)\\
\lambda_2'(\pi) &= -\frac{\partial H}{\partial \sigma} = (1 + (\lambda_a - \lambda_i))f(\pi) - \eta_2(\pi)
\end{aligned}$$

- Complementarity slackness conditions

$$\begin{aligned}
&\lambda_a \geq 0, \lambda_i \geq 0, \eta_1(\pi) \geq 0, \eta_2(\pi) \geq 0 \\
&\lambda_a \Big( B_a - \int_{\pi_{\min}}^{1} \sigma(\pi) f(\pi)\, d\pi \Big) = 0 \\
&\lambda_i \Big( B_i - \int_{\pi_{\min}}^{1} (I - \sigma(\pi)) f(\pi)\, d\pi \Big) = 0 \\
&\eta_1(\pi) u(\pi) = 0 \\
&\eta_2(\pi)(\pi - I - \rho(\pi)\pi + \sigma(\pi)) = 0
\end{aligned}$$

- Transversality conditions

$$\lambda_1(\pi_{\min}) = 0, \lambda_2(\pi_{\min}) = 0, \lambda_1(1) = 0, \lambda_2(1) = 0$$

**Step 2:** Rearranging the optimality condition and establishing $\lambda_a > \lambda_i$

From the optimality condition on $\sigma$ in the Hamiltonian, maximizing $H$ with respect to the allocation between $\sigma$ and $I - \sigma$ yields the first-order condition $\lambda_a - \lambda_i = \frac{\int_{\pi_{\min}}^{1} (\pi - \pi_{\min}) f(\pi)\, d\pi}{\pi_{\min} \int_{\pi_{\min}}^{1} f(\pi)\, d\pi} > 0$. The inequality holds because $\pi > \pi_{\min}$ on the integration domain and under $(A2)$ the agency cannot self-finance. Hence $\lambda_a > \lambda_i$.

Now, from the optimality and transversality conditions:

$$\begin{aligned}
-\eta_1(\pi) &= \lambda_1(\pi) + \lambda_2(\pi)\pi \\
&= \int_{\pi_{\min}}^{\pi} \lambda_1'(v)\, dv + \Big( \int_{\pi_{\min}}^{\pi} \lambda_2'(v)\, dv \Big) \pi \\
&= \int_{\pi_{\min}}^{\pi} \big( \eta_2(v) v - v f(v) + \pi((1 + (\lambda_a - \lambda_i)) f(v) - \eta_2(v)) \big)\, dv \\
&= \int_{\pi_{\min}}^{\pi} (\eta_2(v) v - \pi \eta_2(v))\, dv - \int_{\pi_{\min}}^{\pi} v f(v)\, dv \\
&\quad + (1 + (\lambda_a - \lambda_i)) \pi (F(\pi) - F(\pi_{\min}))
\end{aligned}$$

By the *Fundamental Theorem of Calculus*:

$$\begin{aligned}
\pi(F(\pi) - F(\pi_{\min})) &= \int_{\pi_{\min}}^{\pi} \frac{d(v(F(v) - F(\pi_{\min})))}{dv}\, dv \\
&= \int_{\pi_{\min}}^{\pi} (v f(v) + (F(v) - F(\pi_{\min})))\, dv
\end{aligned}$$

Thus:

$$\begin{aligned}-\eta_1(\pi) &= \int_{\pi_{\min}}^{\pi} \eta_2(v)(v-\pi)\,dv - \int_{\pi_{\min}}^{\pi} vf(v)\,dv \\ &\quad + (1+(\lambda_a-\lambda_i))\int_{\pi_{\min}}^{\pi} (vf(v)+(F(v)-F(\pi_{\min})))\,dv \\ &= \int_{\pi_{\min}}^{\pi} \eta_2(v)(v-\pi)\,dv + (1+(\lambda_a-\lambda_i))\int_{\pi_{\min}}^{\pi} (F(v)-F(\pi_{\min}))\,dv \\ &\quad + (\lambda_a-\lambda_i)\int_{\pi_{\min}}^{\pi} vf(v)\,dv\end{aligned}$$

**Step 3:** Showing $u = 0$

Suppose there exists an interval $\mathcal{I}$ on which $u(\pi) > 0$. By complementarity, $\eta_1(\pi) = 0$ on $\mathcal{I}$, so the expression derived in Step 2 gives:

$$\begin{aligned}&\int_{\pi_{\min}}^{\pi} \eta_2(v)(v-\pi)\,dv + (1+(\lambda_a-\lambda_i))\int_{\pi_{\min}}^{\pi} (F(v)-F(\pi_{\min}))\,dv \\ &\quad + (\lambda_a-\lambda_i)\int_{\pi_{\min}}^{\pi} vf(v)\,dv = 0.\end{aligned}$$

Differentiating twice with respect to $\pi$ yields $\eta_2(\pi) = (1+2(\lambda_a-\lambda_i))f(\pi) + (\lambda_a - \lambda_i)\pi f'(\pi)$.

We now use the *transversality conditions* at $\pi = 1$: $\lambda_1(1) = \lambda_2(1) = 0$. Integrating the multiplier equations *from* $\pi$ *to* 1:

$$\lambda_1(\pi) = \int_{\pi}^{1} (vf(v) - \eta_2(v)v)\,dv, \qquad \lambda_2(\pi) = \int_{\pi}^{1} (\eta_2(v) - (1+(\lambda_a-\lambda_i))f(v))\,dv.$$

Hence $\eta_1(\pi) = -\lambda_1(\pi) - \lambda_2(\pi)\pi = \int_{\pi}^{1} \eta_2(v)(v-\pi)\,dv + (1+(\lambda_a-\lambda_i))\pi\int_{\pi}^{1} f(v)\,dv - \int_{\pi}^{1} vf(v)\,dv \geq 0$ for $\pi < 1$, consistently with the complementarity condition.

**Case A:** $\eta_2(\pi) = 0$ **on a sub-interval** $[\pi_0, \pi_1] \subset \mathcal{I}$**.** Then the expression $\eta_2(\pi) = (1+2(\lambda_a-\lambda_i))f(\pi) + (\lambda_a-\lambda_i)\pi f'(\pi) = 0$ is an ODE whose solution is $f(\pi) = C\pi^{-(2+1/(\lambda_a-\lambda_i))}$ on $[\pi_0, \pi_1]$. Since $\lambda_a > \lambda_i$ (established in Step 2 above), the exponent satisfies $-(2+1/(\lambda_a-\lambda_i)) < -3$. A log-concave density $f$ satisfies $(\log f)'' \leq 0$, i.e. $f''f - (f')^2 \leq 0$. For $f(\pi) = C\pi^{\gamma}$ with $\gamma < -3$, we have $(\log f)'' = -\gamma/\pi^2 > 0$, which contradicts the log-concavity of $f$ in (A1).

**Case B:** $\eta_2(\pi) > 0$ **on** $\mathcal{I}$**.** By complementarity, $(1-\rho(\pi))\pi - (I - \sigma(\pi)) = 0$ on $\mathcal{I}$. Differentiating with respect to $\pi$: $-\rho'(\pi)\pi + (1-\rho(\pi)) + \sigma'(\pi) = 0$. Substituting $\sigma'(\pi) = \pi u(\pi) = \pi\rho'(\pi)$ from $(IC_a^{(1)})$ gives $1 - \rho(\pi) = 0$, so $\rho(\pi) = 1$ and $\sigma(\pi) = I$, implying $u(\pi) = \rho'(\pi) = 0$, which contradicts $u > 0$.

In both cases we reach a contradiction. We conclude that $u(\pi) = 0$ for all $\pi \geq \pi_{\min}$, which establishes that $\rho' = \sigma' = 0$, i.e. the optimal allocation is pooling.

**Part II: threshold, investment, and share.**

**Step 1:** The investor's rationality constraint binds pointwise.

Since the agency is profit-maximizing, $(IR_{i,p})$ binds at the margin. Otherwise, there would exist $\rho^{\#}$ with a strictly higher agency share for which the contract remains feasible:

$$1 - \rho^* > \frac{I - \sigma^*}{\pi^*_{\min}} \Rightarrow \exists \rho^{\#} \in \Big(\rho^*, 1 - \frac{I - \sigma^*}{\pi^*_{\min}}\Big),$$
$$\int_{\pi^*_{\min}}^{1} (\rho^* \pi - \sigma^*) f(\pi)\, d\pi < \int_{\pi^*_{\min}}^{1} (\rho^{\#} \pi - \sigma^*) f(\pi)\, d\pi$$

and $(\sigma^*, \rho^{\#})$ satisfies $(BC_a)$, $(BC_i)$, and $(IR_{i,p})$. Contradiction. Thus:

$$\rho^* = 1 - \frac{I - \sigma^*}{\pi^*_{\min}}$$

**Step 2:** Reducing the agency's optimization problem

The agency solves:

$$\max_{\pi_{\min}, \sigma} \int_{\pi_{\min}}^{1} \Big(\Big(1 - \frac{I - \sigma}{\pi_{\min}}\Big)\pi - \sigma\Big) f(\pi)\, d\pi$$
$$= \max_{\pi_{\min}, \sigma} \int_{\pi_{\min}}^{1} \Big(\pi - \frac{I\pi}{\pi_{\min}} + \frac{\sigma\pi}{\pi_{\min}} - \sigma\Big) f(\pi)\, d\pi$$

under $(BC_a)$ and $(BC_i)$.

The Lagrangian is:

$$L = \int_{\pi_{\min}}^{1} \Big(\pi - \frac{I\pi}{\pi_{\min}} + \frac{\sigma\pi}{\pi_{\min}} - \sigma\Big) f(\pi)\, d\pi$$
$$+ \lambda_1 \Big(B_a - \sigma \int_{\pi_{\min}}^{1} f(\pi)\, d\pi\Big) + \lambda_2 \Big(B_i - (I - \sigma) \int_{\pi_{\min}}^{1} f(\pi)\, d\pi\Big)$$

$$\frac{\partial L}{\partial \sigma} = 0 \Leftrightarrow \int_{\pi_{\min}}^{1} \Big(\frac{\pi}{\pi_{\min}} - 1\Big) f(\pi)\, d\pi - (\lambda_1 - \lambda_2) \int_{\pi_{\min}}^{1} f(\pi)\, d\pi = 0$$

$$\frac{\partial L}{\partial \sigma} = 0 \Leftrightarrow \lambda_1 - \lambda_2 = \frac{\int_{\pi_{\min}}^{1} (\pi - \pi_{\min}) f(\pi)\, d\pi}{\pi_{\min} \int_{\pi_{\min}}^{1} f(\pi)\, d\pi} > 0$$

Thus $\lambda_1 > 0$ and $B_a = \sigma^* \int_{\pi^*_{\min}}^1 f(\pi)\, d\pi$.

$$\begin{aligned}
\frac{\partial L}{\partial \pi_{\min}} = 0 \Leftrightarrow\ & -(\pi_{\min} - I) f(\pi_{\min}) + \int_{\pi_{\min}}^1 \pi(\sigma - I) \frac{\partial}{\partial \pi_{\min}} \Big(\frac{1}{\pi_{\min}}\Big) f(\pi)\, d\pi \\
& + \lambda_1 \sigma f(\pi_{\min}) + \lambda_2 (I - \sigma) f(\pi_{\min}) = 0 \\
\Leftrightarrow\ & -(\pi_{\min} - I) f(\pi_{\min}) + (\lambda_1 - \lambda_2)\sigma f(\pi_{\min}) + \lambda_2 I f(\pi_{\min}) \\
& + \frac{I - \sigma}{\pi^2_{\min}} \int_{\pi_{\min}}^1 \pi f(\pi)\, d\pi = 0 \\
\Leftrightarrow (\pi_{\min} - I) f(\pi_{\min}) =\ & \Big(\frac{\int_{\pi_{\min}}^1 (\pi - \pi_{\min}) f(\pi)\, d\pi}{\pi_{\min} \int_{\pi_{\min}}^1 f(\pi)\, d\pi}\Big)\Big(\frac{B_a}{\int_{\pi_{\min}}^1 f(\pi)\, d\pi}\Big) f(\pi_{\min}) \\
& + \lambda_2 I f(\pi_{\min}) + \frac{I - \frac{B_a}{\int_{\pi_{\min}}^1 f(\pi)\, d\pi}}{\pi^2_{\min}} \int_{\pi_{\min}}^1 \pi f(\pi)\, d\pi
\end{aligned}$$

Thus $\pi^*_{\min} > I$.

**Case 1 (cash-rich, under (A2)):** $B_i > (I - \sigma^*) \int_{\pi^*_{\min}}^1 f(\pi)\, d\pi$, i.e. $\pi^*_{\min} > (1 - F)^{-1}(\frac{B_a + B_i}{I})$.

Under assumption $(A2)$, the total budget is sufficient to finance all positive-NPV projects, so the investor's budget constraint $(BC_i)$ is slack and $\lambda_2 = 0$.

The first-order condition on $\pi_{\min}$ becomes:

$$\begin{aligned}
& (\pi_{\min} - I) f(\pi_{\min}) - \Big(\frac{\int_{\pi_{\min}}^1 (\pi - \pi_{\min}) f(\pi)\, d\pi}{\pi_{\min} \int_{\pi_{\min}}^1 f(\pi)\, d\pi}\Big)\Big(\frac{B_a}{\int_{\pi_{\min}}^1 f(\pi)\, d\pi}\Big) f(\pi_{\min}) \\
& - \frac{I - \frac{B_a}{\int_{\pi_{\min}}^1 f(\pi)\, d\pi}}{\pi^2_{\min}} \int_{\pi_{\min}}^1 \pi f(\pi)\, d\pi = 0
\end{aligned}$$

Since the agency's reduced objective $V(\pi_{\min})$ is continuous on the compact set $[I, \bar{p}]$ (where $\bar{p}$ is the largest feasible threshold satisfying $\sigma^* = B_a / \bar{F}(\pi_{\min}) \le I$), it attains a maximum. At any interior maximum $\pi^*_{\min}$, the first-order condition holds. The right-hand side is strictly positive, so $\pi^*_{\min} > I$. The optimal allocation is then $\sigma^* = B_a / \bar{F}(\pi^*_{\min})$ and $\rho^* = 1 - (I - \sigma^*)/\pi^*_{\min}$. If the maximum is at $\bar{p}$, then $\sigma^* = I$, $\rho^* = 1$ and $\pi^*_{\min} = \bar{p} = (1 - F)^{-1}(B_a / I)$, which exceeds $I$ by (A2). No threshold above $\bar{p}$ does better: there $\sigma = I$, $(BC_a)$ is slack, and the agency's payoff $\int_{\pi_{\min}}^1 (\pi - I) f(\pi)\, d\pi$ decreases in $\pi_{\min}$.

**Case 2 (cash-poor boundary):** $\lambda_2 = 0$ and $B_i = (I - \sigma^*) \int_{\pi^*_{\min}}^1 f(\pi)\, d\pi$.

This is the boundary of the cash-poor regime. Here the unconstrained optimum coincides with the budget-constrained threshold: $\pi^*_{\min} = (1-F)^{-1}(\frac{B_a+B_i}{I})$.

**Case 3 (cash-poor):** $\lambda_2 > 0$ and $B_i = (I - \sigma^*)\int_{\pi^*_{\min}}^1 f(\pi)\,d\pi$.

Both budget constraints bind. Thus:

$$\pi^*_{\min} = (1-F)^{-1}\Big(\frac{B_a+B_i}{I}\Big), \quad \sigma^* = \frac{B_a}{B_a+B_i}I$$

The condition $\lambda_2 > 0$ imposes:

$$\begin{aligned}
&(\pi^*_{\min} - I)\, f(\pi^*_{\min}) \\
&- \left(\frac{\int_{\pi^*_{\min}}^1 (\pi - \pi^*_{\min})\, f(\pi)\,d\pi}{\pi^*_{\min}\int_{\pi^*_{\min}}^1 f(\pi)\,d\pi}\right)\left(\frac{B_a}{\int_{\pi^*_{\min}}^1 f(\pi)\,d\pi}\right) f(\pi^*_{\min}) \\
&- \left(\frac{I - \frac{B_a}{\int_{\pi^*_{\min}}^1 f(\pi)\,d\pi}}{(\pi^*_{\min})^2}\right)\int_{\pi^*_{\min}}^1 \pi\, f(\pi)\,d\pi \\
&= \lambda_2\, I\, f(\pi^*_{\min}) \;>\; 0.
\end{aligned}$$

The left-hand side is strictly positive, meaning the budget-constrained threshold $(1-F)^{-1}(\frac{B_a+B_i}{I})$ is *above* the unconstrained optimum from Case 1. With a binding investor budget, the agency finances fewer projects than in Case 1.

## 8.4 Proof of Proposition 4 (cash-poor vs. cash-rich).

The Lagrangian of the ex-ante commitment program (see proof of Proposition 3) has stationarity conditions $\lambda^*_1 = \lambda^*_2$ and $\pi^*_{\min} = I + \lambda^*_1 I$. There are three cases.

**Region (1): severely cash-poor.** When $B_a + B_i < I(1 - F(\pi^{\text{ep}}_{\min}(B_a)))$, both budget constraints bind in both regimes. Hence

$$\pi^*_{\min} = (1-F)^{-1}\Big(\tfrac{B_a+B_i}{I}\Big), \qquad \sigma^* = \tfrac{B_a}{B_a+B_i}I,$$

and $\rho^* = 1 - (I - \sigma^*)/\pi^*_{\min}$ in both regimes. Since the ex-post commitment first-order condition of Proposition 2 would require $\pi^{\text{ep}}_{\min}(B_a)$ strictly below this budget-constrained threshold, the budget dominates and both regimes coincide.

**Region (2): moderately cash-poor.** When $I(1 - F(\pi^{\text{ep}}_{\min}(B_a))) \le B_a + B_i < I(1 - F(I))$, the budget-constrained threshold $\pi^{\text{bdg}}_{\min} := (1-F)^{-1}((B_a+B_i)/I)$ lies in $(I, \pi^{\text{ep}}_{\min}(B_a)]$. Under ex-ante commitment investors, the constrained-efficient allocation finances every project above $\pi^{\text{bdg}}_{\min}$. Under ex-post commitment investors, the

informational friction raises the threshold to $\pi^{\text{ep}}_{\min}(B_a) \geq \pi^{\text{bdg}}_{\min}$, excluding the interval $[\pi^{\text{bdg}}_{\min}, \pi^{\text{ep}}_{\min}(B_a))$.

**Region (3): cash-rich.** When $B_a + B_i \geq I(1 - F(I))$, which is the second part of (A2), the ex-ante commitment threshold drops to $\pi^{*}_{\min} = I$ (Proposition 3), while the ex-post commitment threshold remains $\pi^{\text{ep}}_{\min}(B_a) > I$ (Proposition 2). The two regimes diverge on the interval $[I, \pi^{\text{ep}}_{\min}(B_a))$. □

### 8.5 Uniqueness of the ex-post commitment threshold under $f$ uniform

For $f \equiv 1$ on $[0,1]$: $1 - F(p) = 1 - p$ and $\int_p^1 \pi \, d\pi = (1 - p^2)/2$. Substituting $\sigma^* = B_a/(1-p)$ and $\rho^* = 1 - (I - \sigma^*)/p$ into the agency's payoff $\int_p^1 (\rho^* \pi - \sigma^*) f(\pi) \, d\pi$ gives

$$V(p) = \frac{1-p}{2p}\big[p^2 + p(1-I) - I + B_a\big].$$

Differentiating,

$$V'(p) = \frac{-2p^3 + Ip^2 + I - B_a}{2p^2}, \qquad V''(p) = -\frac{p^3 + I - B_a}{p^3}.$$

Since $p > I > 0$ and $B_a < I$ from (A2), $p^3 + I - B_a > 0$, so $V''(p) < 0$ for all $p > I$. The objective is strictly concave. Setting $V'(p) = 0$ gives the unique stationary point as the solution of $2p^3 - Ip^2 - (I - B_a) = 0$ in $(I, 1)$. The constraint $\sigma^* \leq I$ reads $p \leq \bar{p} = 1 - B_a/I$. At $p = \bar{p}$ the cubic factors as $p\,(2p^2 - Ip - I)$, so the stationary point lies below $\bar{p}$ if and only if $B_a < \hat{B}_a := I(1 - \hat{p})$, with $\hat{p} = \big(I + \sqrt{I^2 + 8I}\big)/4$. The optimal threshold is the stationary point for $B_a < \hat{B}_a$ and $\bar{p} = 1 - B_a/I$ for $B_a \geq \hat{B}_a$. □

# References


Axelson, U., Strömberg, P., and Weisbach, M. (2009). Why are buyouts levered? The financial structure of private equity funds. *Journal of Finance*, 64(4):1549–1582.

Casamatta, C. (2003). Financing and advising: Optimal financial contracts with venture capitalists. *Journal of Finance*, 58(5):2059–2085.

Cohen, S. (2013). What do accelerators do? insights from incubators and angels. *Innovations: Technology, Governance, Globalization*, 8(3–4):19–25.

Cohen, S. and Hochberg, Y. V. (2014). Accelerating startups: The seed accelerator phenomenon. Working Paper 2418000, SSRN.

Dossou, S. (2026). Protection and dilution in sequential venture financing. Working paper, Mines Paris – PSL.

Ewens, M., Nanda, R., and Rhodes-Kropf, M. (2018). Cost of experimentation and the evolution of venture capital. *Journal of Financial Economics*, 128(3):422–442.

Fang, L., Ivashina, V., and Lerner, J. (2015). The disintermediation of financial markets. *Journal of Financial Economics*.

Gonzalez-Uribe, J. and Leatherbee, M. (2017). The effects of business accelerators on venture performance: Evidence from start-up chile. *The Review of Financial Studies*.

Gourier, E., Phalippou, L., and Westerfield, M. M. (2024). Capital commitment. *The Journal of Finance*, 79(5):3407–3457.

Guesnerie, R. and Laffont, J.-J. (1984). A complete solution to a class of principal-agent problems. *Journal of Public Economics*.

Hochberg, Y. V. (2016). Accelerating entrepreneurs and ecosystems: The seed accelerator model. *Innovation Policy and the Economy*, 16:25–51.

Holmstrom, B. and Myerson, R. B. (1983). Efficient and durable decision rules with incomplete information. *Econometrica*, 51(6):1799–1819.

Howell, S. T. (2017). Financing innovation: Evidence from r&d grants. *American Economic Review*, 107(4):1136–1164.

Kamien, M. and Schwartz, N. (1991). *Dynamic Optimization*. North-Holland.

Kim, J.-H. and Wagman, L. (2014). Portfolio size and information disclosure: An analysis of startup accelerators. *Journal of Corporate Finance*, 29:520–534.

Lerner, J., Mao, J., Schoar, A., and Zhang, N. R. (2022). Investing outside the box: Evidence from alternative vehicles in private equity. *Journal of Financial Economics*, 143(1):359–380.

Lerner, J. and Nanda, R. (2020). Venture capital's role in financing innovation: What we know and how much we still need to learn. *Journal of Economic Perspectives*, 34(3):237–261.

Maskin, E. and Tirole, J. (1992). The principal-agent relationship with an informed principal. *Econometrica*, 60(1):1–42.

Metrick, A. and Yasuda, A. (2010). The economics of private equity funds. *The Review of Financial Studies*, 23(6):2303–2341.

Myerson, R. B. (1983). Mechanism design by an informed principal. *Econometrica*, 51(6):1767–1797.

Zarei, H., Rasti-Barzoki, M., and Moon, I. (2022). A game theoretic approach to the selection, mentorship, and investment decisions of start-up accelerators. *IEEE Transactions on Engineering Management*, 69(4):1753–1768.

Zeisberger, C., Prahl, M., and White, B. (2017). *Mastering Private Equity*. Wiley.